

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Fall

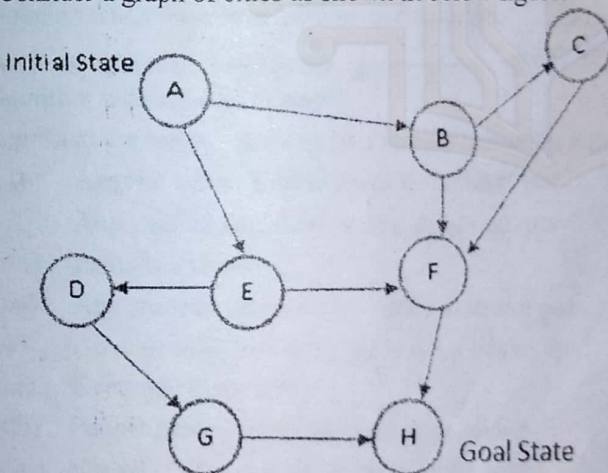
Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is Artificial Intelligence? Discuss the impact of artificial intelligence on society. 7
- b) What is an intelligent agent? Describe the task environment of an autonomous car. 8
- a) What are the components of a state space search problem formulation? Formulate the 8-queens problem. 7 2
- b) How does the interactive deepening search resolve the problem of depth-first search and breath- first search? Explain with an example. 8
- a) Consider a graph of cities as shown in below figure: 7



- Represent state space of this graph using prolog.
- Write a simple search algorithm in prolog to reach to the goal state from initial state.

- b) Represent the following facts in predicate logic:
 - i. John likes all kind of food.
 - ii. Apple and vegetable are food.
 - iii. Anything anyone eats and not killed is food.
 - iv. Anil eats peanuts and still alive.
 - v. Harry eats everything that Anil eats.
- 4. a) Consider the facts given in Q.3 b) using resolution prove that John likes peanuts.
- b) What is machine learning? Discuss the difference between regression, classification and clustering in machine learning.
- 5. a) Discuss Naive Bayes' algorithm with an example.
- b) How can we find the right hyperplane in SVM? Using K-means clustering, cluster the following data into two clusters and show each step.
{2, 4, 10, 12, 3, 20, 30, 11, 25}.
- 6. a) Draw the architecture of Expert System. How forward chaining and backward chaining assist the inference engine of Expert System?
- b) What are the different types of ambiguities that occur in Natural Language Processing? Explain with examples.
- 7. Write short notes on: (Any two)
 - a) Fuzzy learning
 - b) Best first search
 - c) Hopfield Network